

Upfront

913

The number of "neurons" in a brainlike chip made by IBM using a powerful new memory material.



Google's Intelligence Designer

The man behind a startup Google acquired for \$650 million plans to build a revolutionary new artificial intelligence.

By Tom Simonite

Demis Hassabis started playing chess at age four and soon blossomed into a child prodigy. At age eight, success on the chessboard led him to ponder two questions that have obsessed him ever since: first, how does the brain learn to master complex tasks; and second, could computers ever do the same?

Now 38, Hassabis puzzles over those questions for Google, having sold his little-known London-based startup, DeepMind Technologies, to the search company last year for a reported 400 mil-

lion pounds (\$650 million at the time). Google snapped up DeepMind shortly after it demonstrated software capable of teaching itself to play classic video games at a superhuman level. At the TED conference in Vancouver this year, Google CEO Larry Page gushed about Hassabis and called his company's technology "one of the most exciting things I've seen in a long time."

DeepMind's technology could improve some of Google's existing products. But if the technology progresses, it could change the role computers play in many fields. Building software that can learn when faced with almost any problem could help address some of the world's most intractable problems, says Hassabis. "AI has huge potential to be amazing for humanity," he says. "It will really accelerate progress in solving disease and all these things we're

making relatively slow progress on at the moment."

Hassabis's quest to understand and create intelligence has led him through three careers: game developer, neuroscientist, and now artificial-intelligence entrepreneur. At 17, he led development of the classic simulation game Theme Park, released in 1994. He went on to complete a degree in computer science at the University of Cambridge and founded his own successful games company in 1998. But the demands of building successful computer games limited the time Hassabis could devote to his true calling. "I thought it was time to do something that focused on intelligence," he says.

So in 2005, Hassabis began a PhD in neuroscience at University College London, with the idea that studying real brains might turn up clues that could help with artificial intelligence. He chose to study the hippocampus, a part of the brain that underpins memory and spatial navigation, which is still relatively poorly understood.

He soon made a mark. In a 2007 study recognized by the journal *Science* as a "Breakthrough of the Year," he showed that five patients suffering amnesia due to damage to the hippocampus struggled to imagine future events. It suggested that a part of the brain thought to be concerned only with the past is also crucial to planning for the future.

That memory and forward planning are intertwined was one idea Hassabis took with him into his next venture. In 2011, he quit life as a postdoctoral researcher to found DeepMind, whose stated goal was to "solve intelligence." The company hired leading researchers in machine learning and attracted noteworthy investors, including Peter Thiel's firm Founders Fund and Elon Musk, founder of Tesla and SpaceX. But DeepMind kept a low profile until December

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2013, when it staged a kind of debutante moment at a leading research conference on machine learning. At Harrah's Casino on the shore of Lake Tahoe, DeepMind researchers showed off software that had learned to play three classic Atari games—Pong, Breakout, and Enduro—better than a human expert. The software was equipped only with access to the controls and display, a knowledge of the score, and an instinct to make that score as high as possible. It became an expert gamer entirely through trial and error.

DeepMind made use of a newly fashionable machine-learning technique called deep learning, which involves processing data through networks of crudely simulated neurons. But it combined deep learning with other tricks to make something with an unexpected level of intelligence.

"People were a bit shocked because they didn't expect that we would be able to do that at this stage of the technology," says Stuart Russell, a professor and artificial-intelligence specialist at the University of California, Berkeley.

In particular, DeepMind had used a technique called reinforcement learning, which is inspired by the work of animal psychologists such as B. F. Skinner. This led to software that learns by taking actions and receiving feedback on the results of those actions. Artificial-intelligence researchers have been tin-

kering with reinforcement learning for decades. But until DeepMind's Atari demo, no one had built a system capable of learning anything nearly as complex as how to play a computer game. One reason it was possible was a trick borrowed from Hassabis's favorite area of the brain. Part of the Atari-playing software's learning process involved replaying its past experiences over and over to try to extract the most accurate hints on what it should do in the future. "That's something that we know the brain does," says Hassabis.

Hassabis expects swift progress toward a very powerful new form of artificial intelligence.

"When you go to sleep, your hippocampus replays the memory of the day back to your cortex."

More than a year later, Russell and other researchers are still puzzling over exactly how that trick and others led to such remarkable results, and what else they might be used for. But Google didn't take long to recognize the importance of the technology, announcing a month after the Tahoe demonstration that it had acquired the company.

Today Hassabis leads what is now called Google DeepMind. The group was

roughly 75 people strong at the time it joined Google, and around three-quarters of its members work on fundamental research. The rest form an "applied research team" that looks for opportunities to apply DeepMind's techniques to existing Google products. In addition, Hassabis dreams of creating "AI scientists" that could do things like generate and test new hypotheses about disease in the lab. When prodded, he says that DeepMind's software could also be useful to robotics, an area in which Google has recently invested heavily. "One reason we don't have more robots doing more helpful things is that they're usually preprogrammed," he says. "They're very bad at dealing with the unexpected or learning new things."

Hassabis's reluctance to talk about applications might be coyness, or it could be that his researchers are still in the early stages of understanding how to advance the company's AI software. One strong indicator that Hassabis expects swift progress toward a powerful new form of AI is that he is setting up an ethics board to consider the possible downsides of advanced artificial intelligence. "It's something that we or other people at Google need to be cognizant of. We're still playing Atari games currently," he says, laughing. "But we are on the first rungs of the ladder."

TO MARKET

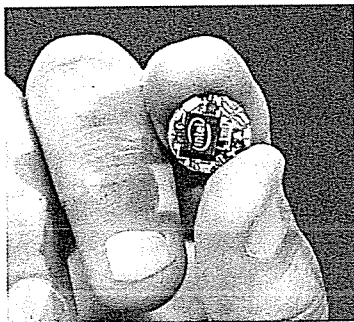
On Button

Curie

COMPANY:
Intel

PRICE:
Not disclosed

AVAILABILITY:
Second half of 2015



Intel has developed a minuscule computer, called Curie, to power compact wearable devices. About the size of a button, Curie includes a Bluetooth radio, a six-axis motion sensor with accelerometer and gyroscope, and components that can rapidly and precisely differentiate between different types of physical activity. The device runs on a coin-size battery (although Intel hasn't said for how long). Intel's announcement is a sure sign that hardware makers are eager to build wearable devices of all kinds. It also points to the unwelcome size and bulk of many existing smart watches and smart glasses, a factor that has turned many would-be users away.